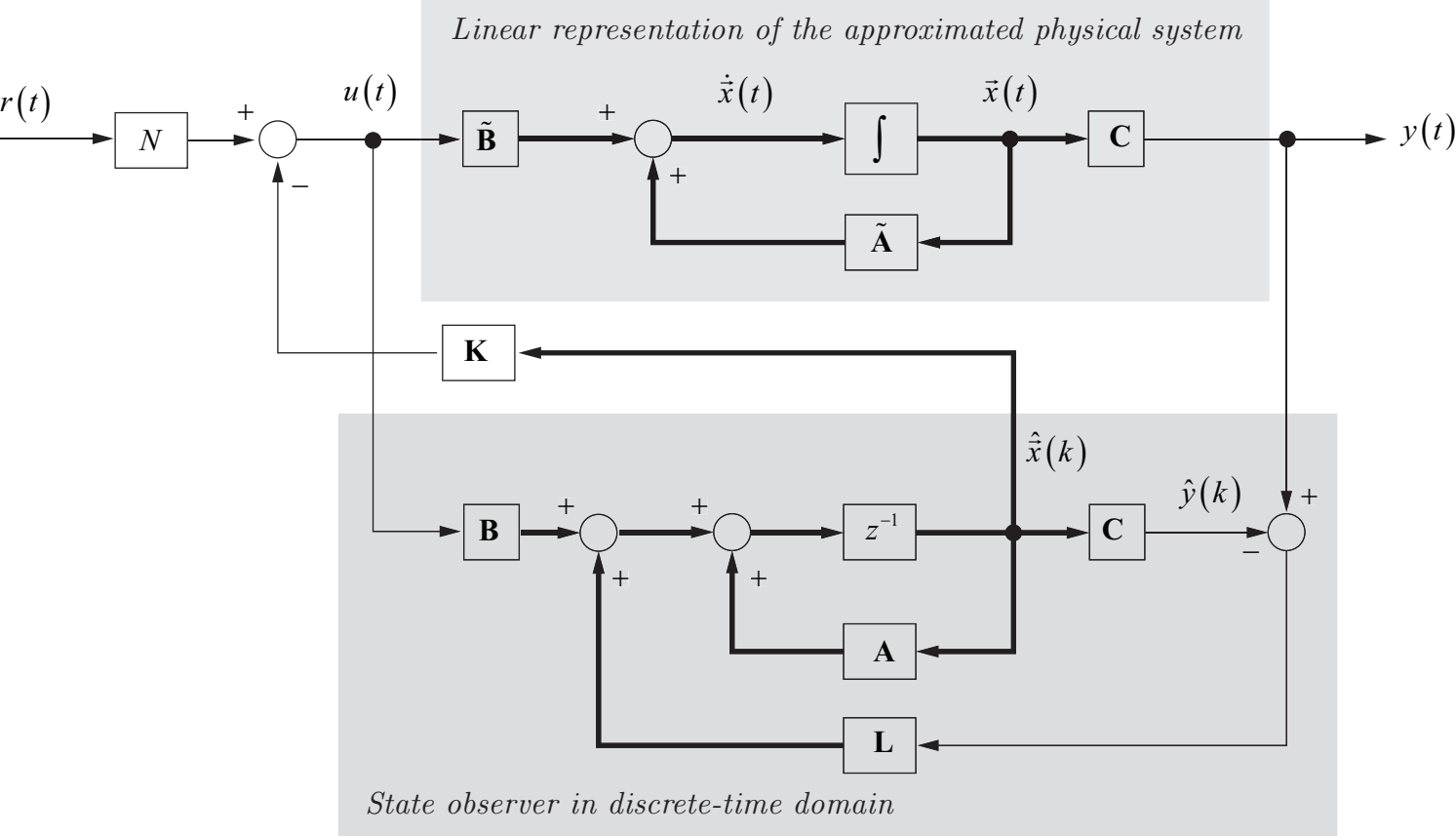
Linear representation of the approximated physical system
r(t)
N
+
−
u(t)
B̃
+
+
ẋ̃(t)
∫
x̃(t)
C
y(t)
+
Ã
K
x̂̇(k)
ŷ(k)
+
B
+
+
z⁻¹
C
−
+
+
A
L
State observer in discrete-time domain